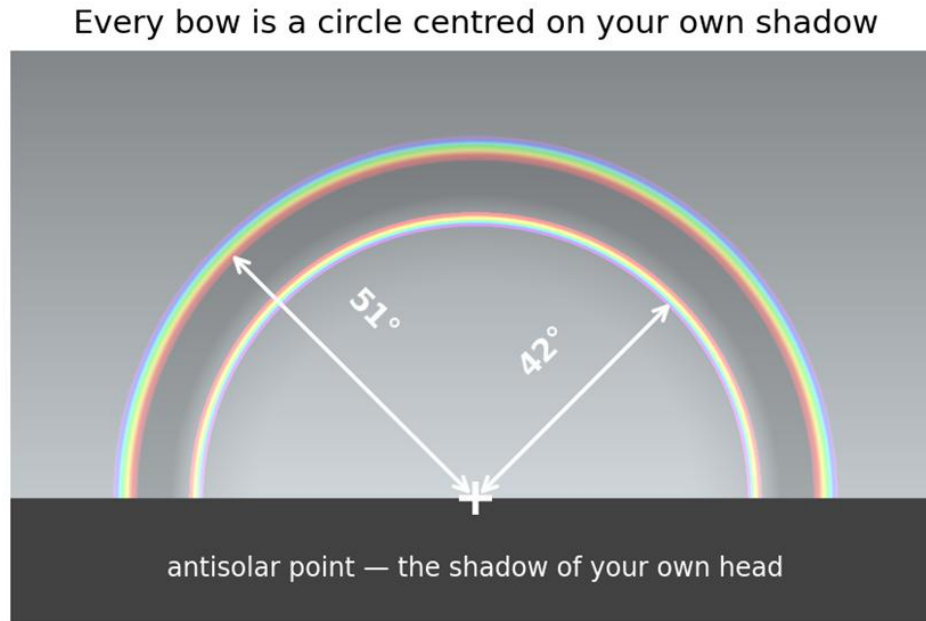


Seventeen Questions

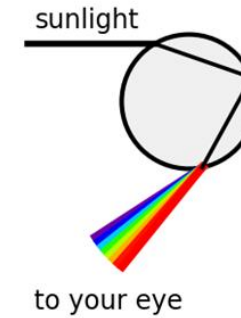
Everyday things, and what is actually going on underneath

Unofficial Club

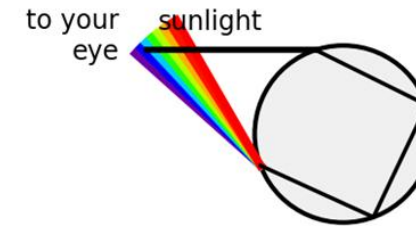
Why is a rainbow an arc — and why is there often a second one?



one bounce inside → light comes back at 42°



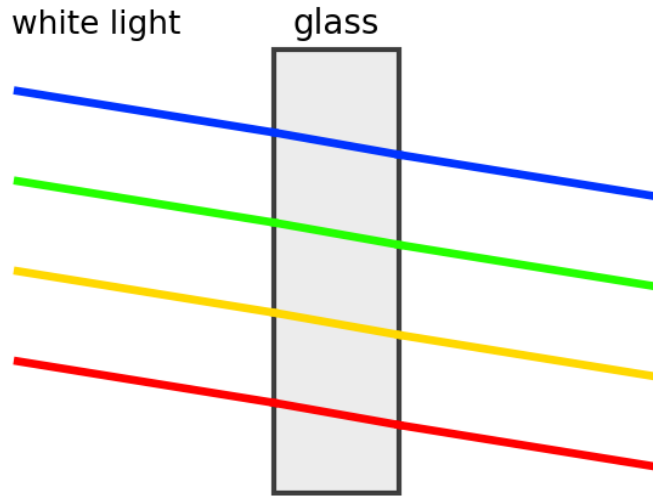
two bounces inside → light comes back at 51°



Every raindrop sends sunlight back at one fixed angle: about 42° away from the point directly opposite the sun. All the drops sitting at that angle form a circle centred on your own shadow, and the ground cuts off the bottom of it — so you see an arc. The second bow is light that bounced twice inside the drops instead of once, which returns it at 51° and flips the colour order.

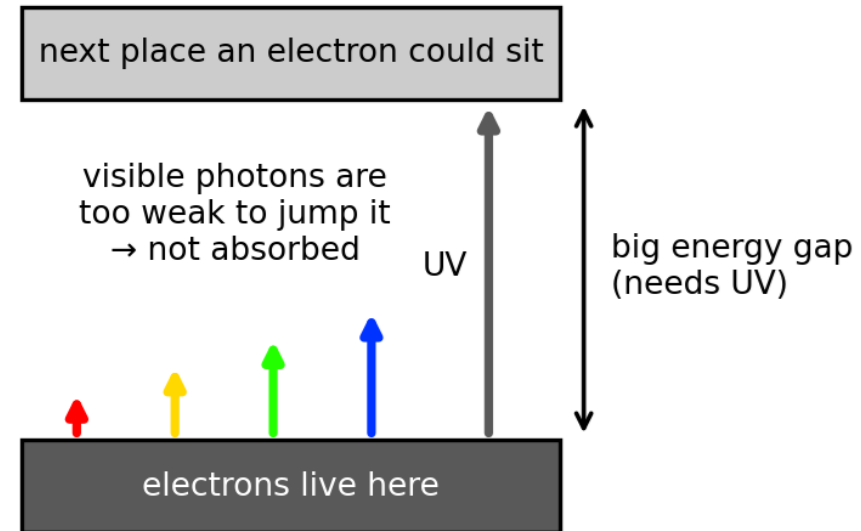
Why is glass transparent? Why can light go through it?

Light goes straight through



nothing in glass can take the light's energy,
so it just keeps going

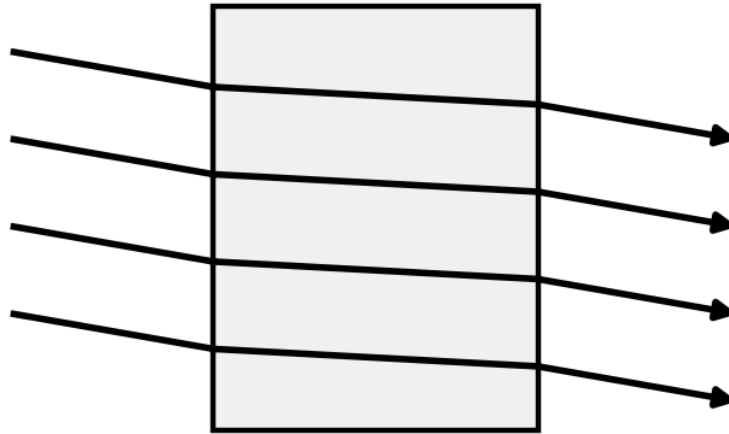
The reason: nothing to absorb it with



A material can only absorb a photon whose energy exactly matches a step its electrons are allowed to climb. In glass the smallest step available is so large that only ultraviolet can make the jump, so visible light has nothing to hand its energy to and passes straight through. The same fact is why a window still blocks much of the sun's UV.

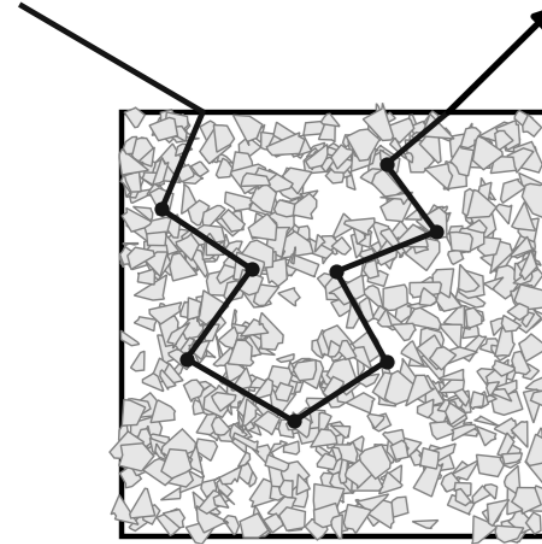
Ice and snow are the same substance — why do they look so different?

solid ice — only 2 surfaces



light passes almost straight through
→ looks clear

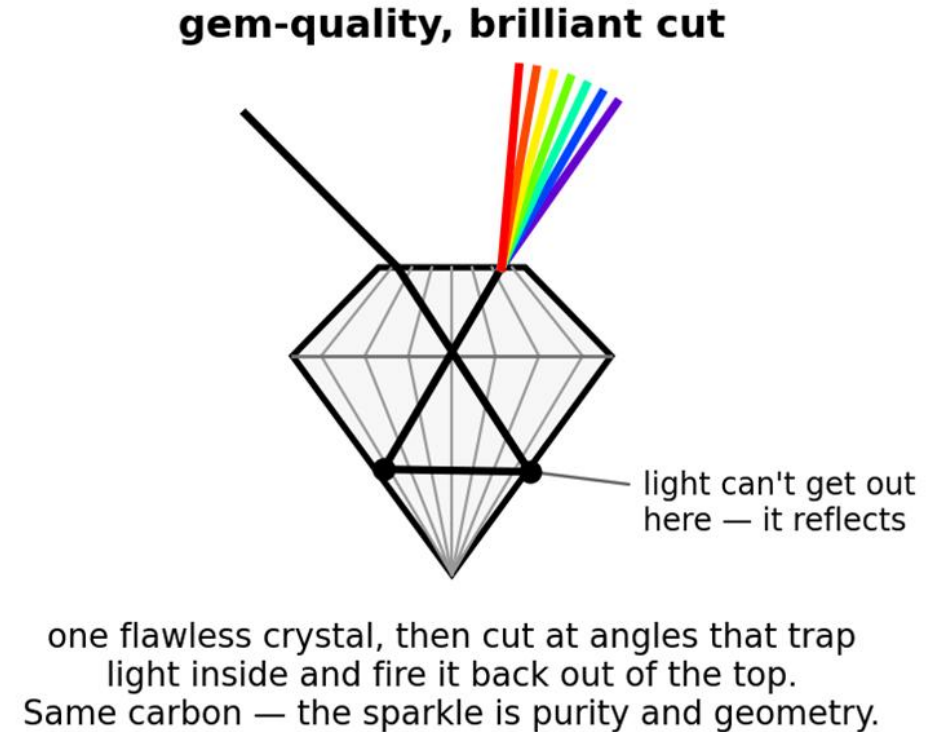
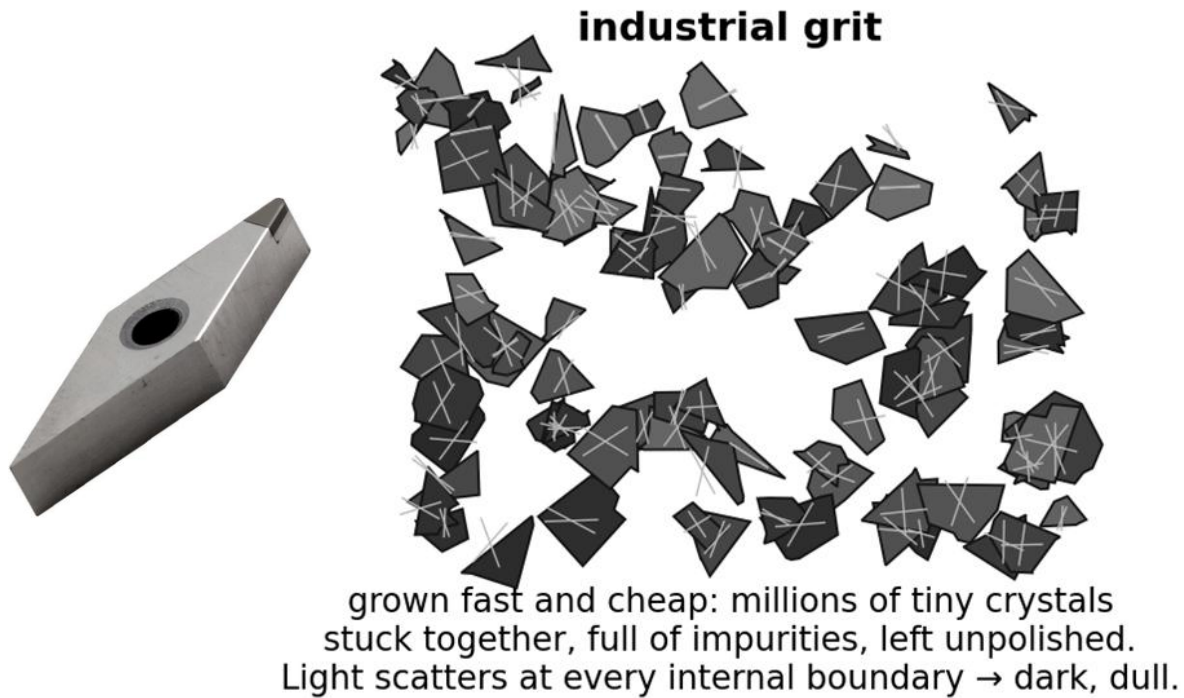
snow — millions of tiny crystals



light bounces at every grain boundary and comes
back out — all colours equally → looks white

Ice is one solid block with just two surfaces, so light travels almost straight through and you see whatever is behind it. Snow is millions of tiny crystals with air in between, and light bounces at every single boundary until it scatters back out the way it came. Because every colour scatters equally well, what comes back is white.

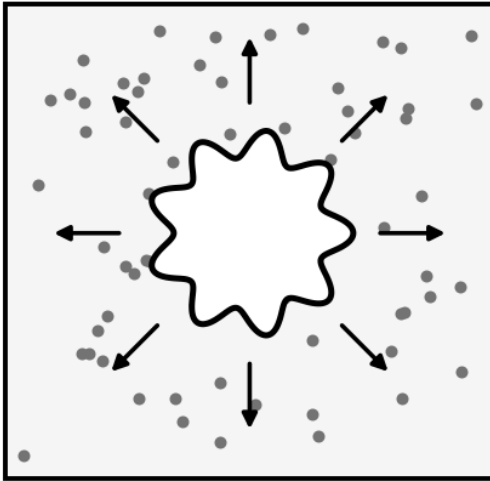
Why does industrial cutting diamond look nothing like jewellery diamond?



Industrial diamond is grown fast and cheap as a mass of tiny crystals packed together, loaded with impurities and never polished — light scatters at all those internal boundaries, so it looks dark and dull. A gem is a single flawless crystal, cut at angles chosen to trap light inside and fire it back out through the top. Same carbon; the difference is purity and geometry.

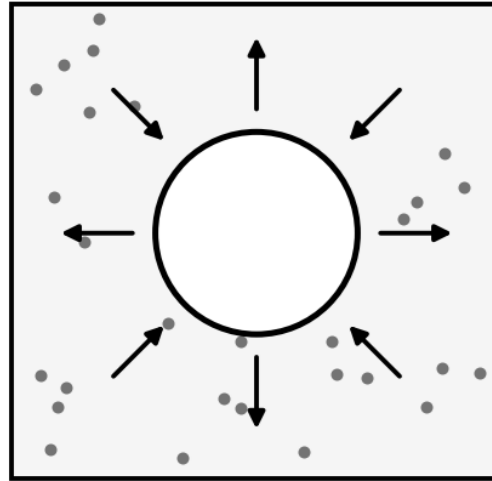
What would pure water do to your body?

salty water
(saltier than the cell)



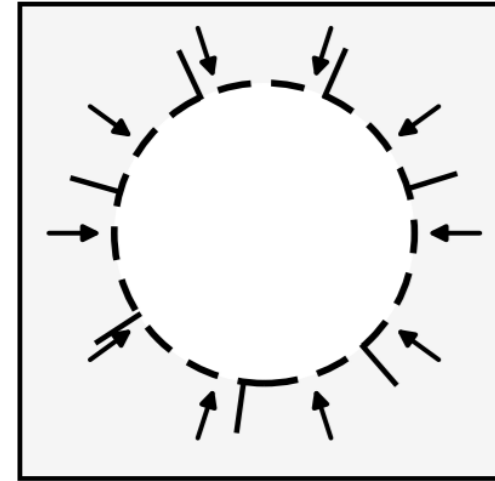
water leaves
cell shrivels

normal body fluid
(balanced)

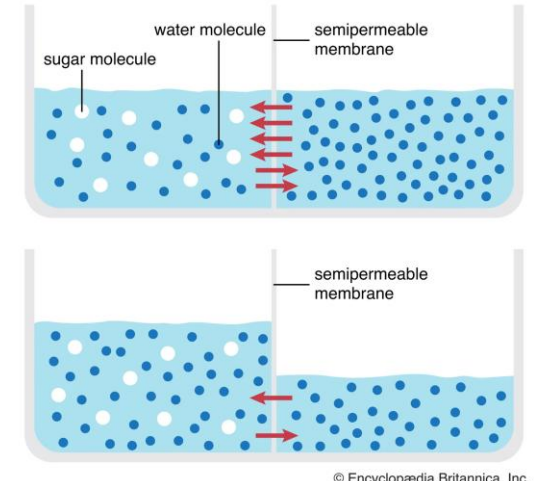


water in = water out
cell is happy

pure water
(no salt at all)



water floods in
cell swells and bursts

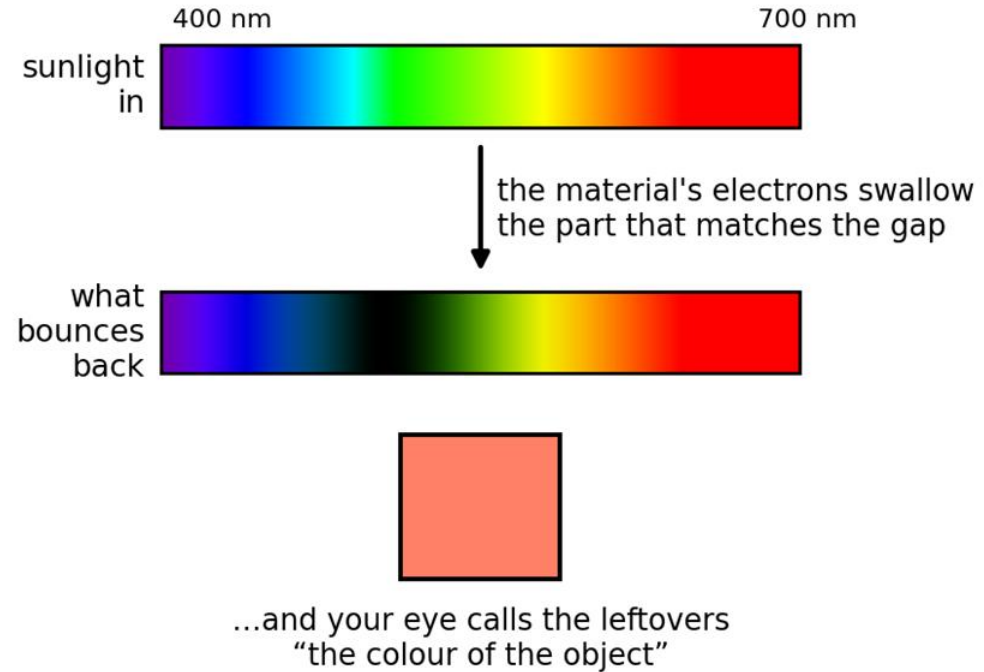
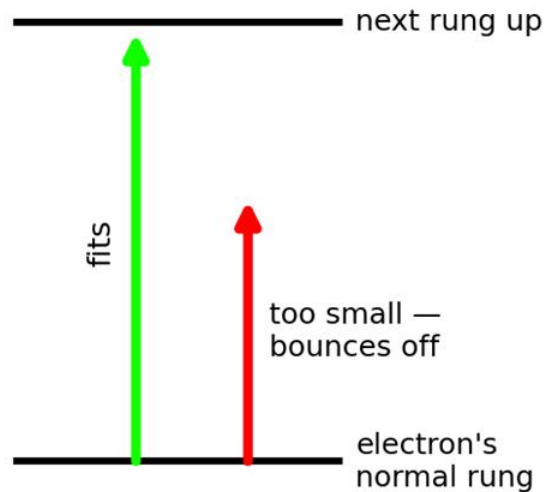


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A glass of it is harmless — your stomach immediately mixes it with everything else in there. The danger is volume, because your cells are salty inside and water always flows toward the saltier side: surrounded by pure water, it floods in and cells swell. Drink litres of it quickly, or put it straight into a vein, and you dilute the salt in your blood badly enough to be dangerous.

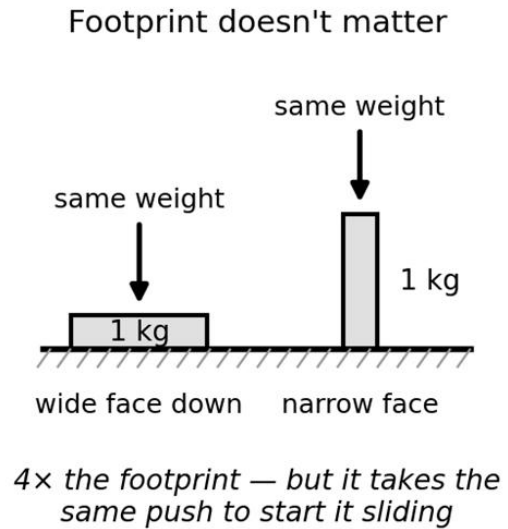
What is colour, really?

A colour is an energy gap

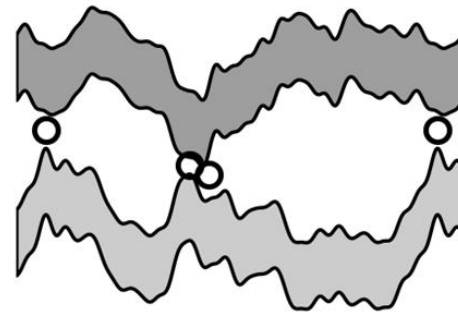


Electrons in a material can only sit on certain energy rungs, and the size of the gaps between those rungs is set by the material's chemistry. A photon gets absorbed only if its energy matches a gap; if nothing matches, it is sent back out. So the colour you see is the leftovers — the part of the light the material had no gap to swallow.

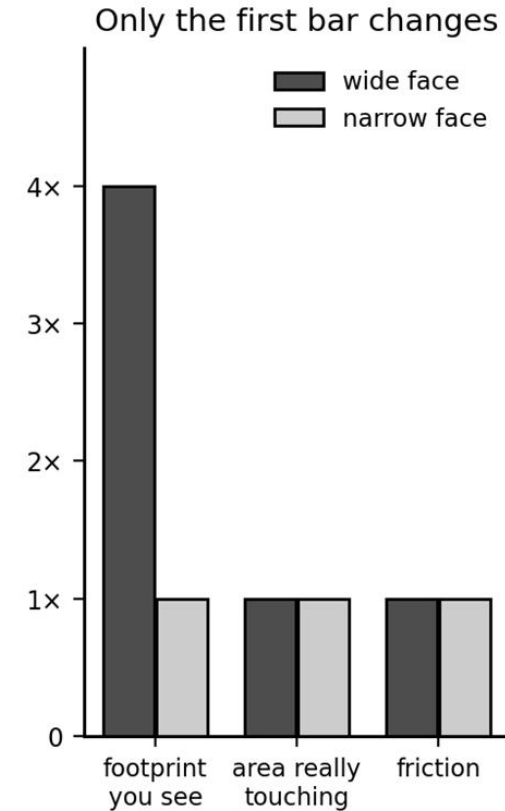
Why doesn't friction depend on contact area?



zoom in far enough: the surfaces only touch at a handful of high spots

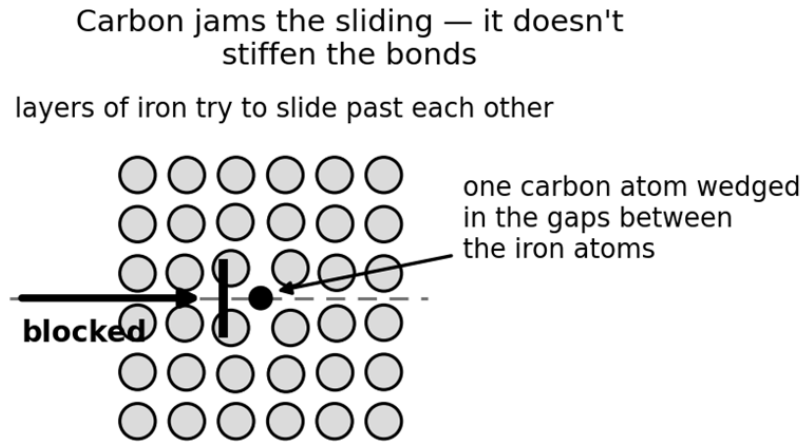


press harder and those spots squash and grow. Real contact follows the weight, not the footprint.



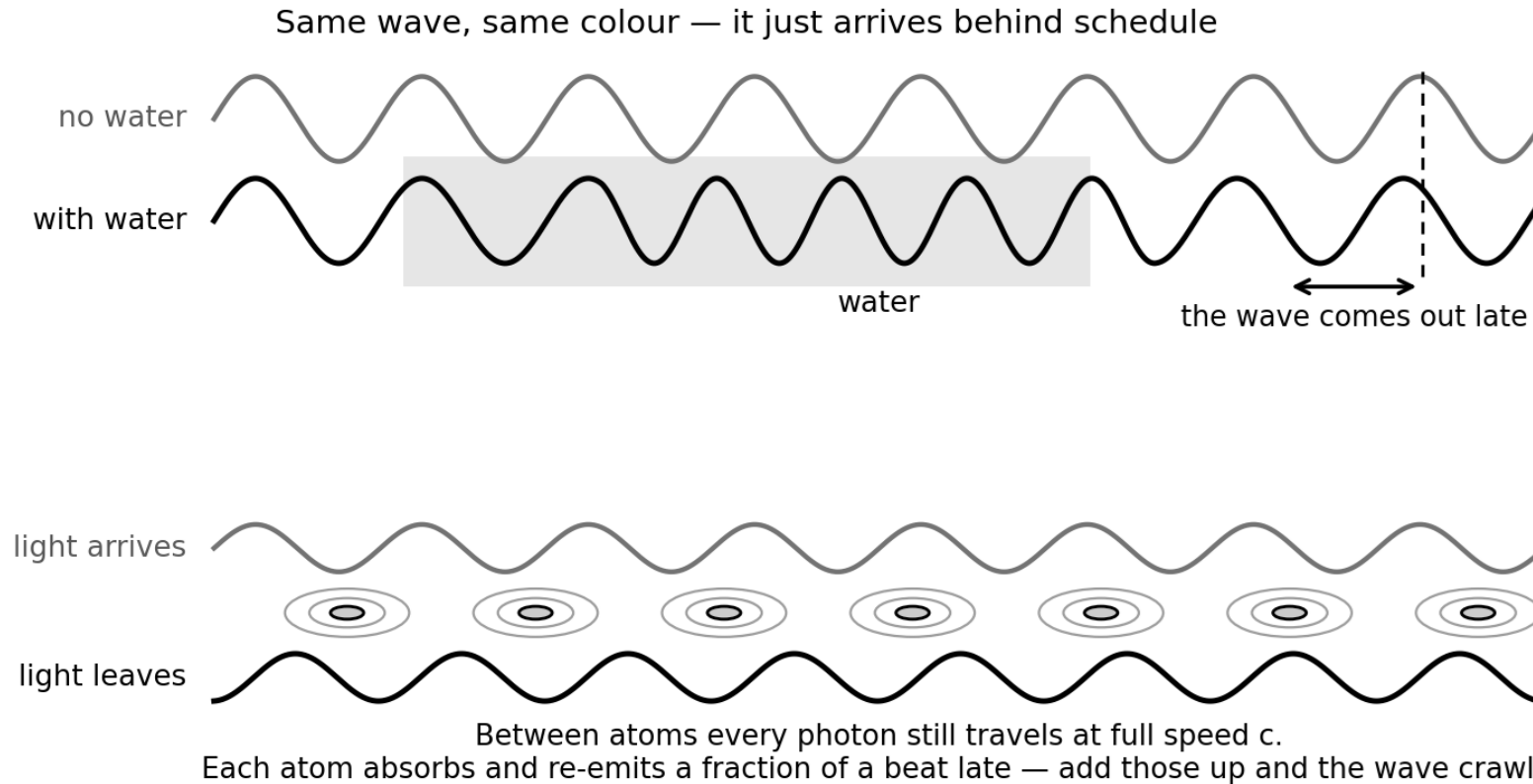
No surface is really flat: two objects touch only at a scattering of microscopic high spots, and the area genuinely in contact is far smaller than the footprint you can see. Add weight and those spots squash and multiply, so real contact tracks the weight rather than the footprint. Spread the same weight over a larger face and each spot simply carries less — and it cancels out exactly.

How does a few percent of carbon change iron so much?



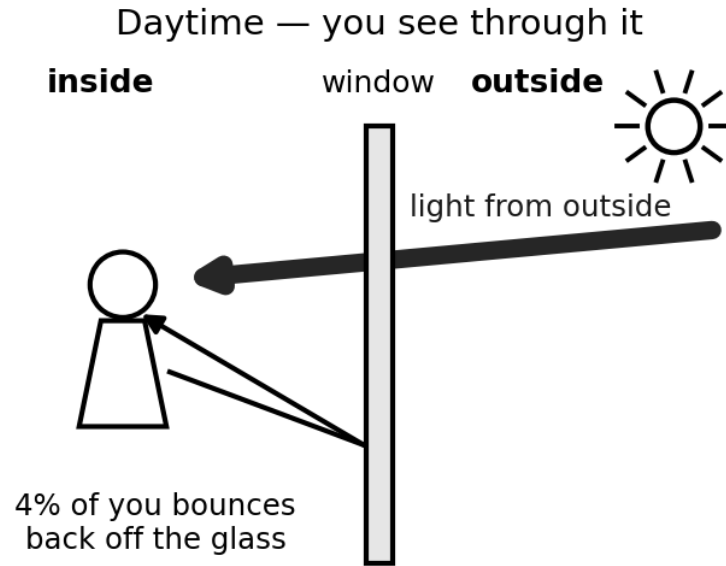
Here is the twist — carbon barely changes the stiffness: pure iron and steel both sit near 200 GPa, because Young's modulus comes from the iron–iron bonds, which carbon never touches. What changes enormously is strength and hardness, because iron gives way when planes of atoms slide over one another, and carbon wedged into the gaps (plus the hard carbide particles it forms) jams that sliding. At about 2% carbon you have left steel behind entirely: that is cast iron — hard and wear-resistant, but brittle.

If the speed of light can't change, why do we say it slows down in water?

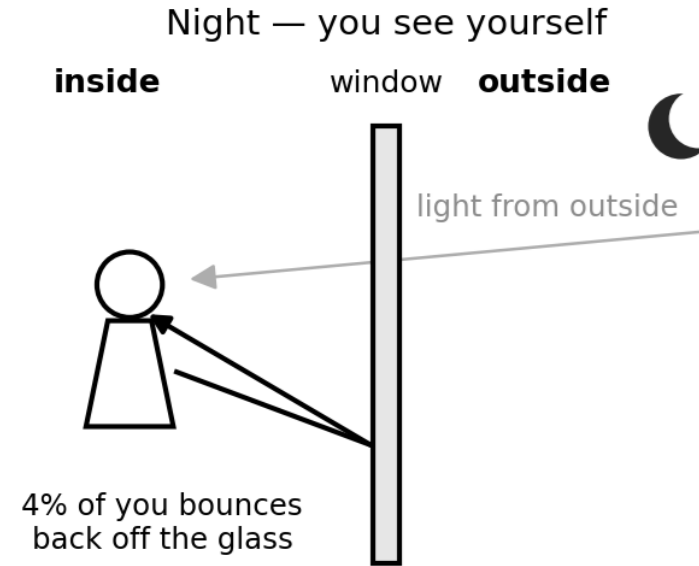


It doesn't — between atoms, every photon still travels at exactly c . What slows down is the wave you would actually measure: each atom absorbs the light and re-radiates it a fraction of a beat late, and those tiny delays pile up over trillions of atoms. The resulting wave crawls through at about three quarters of c , and that lag is what bends the beam.

Why can you see yourself in a window at night, but see through it by day?



the view outside is thousands of times brighter,
so it drowns your faint reflection

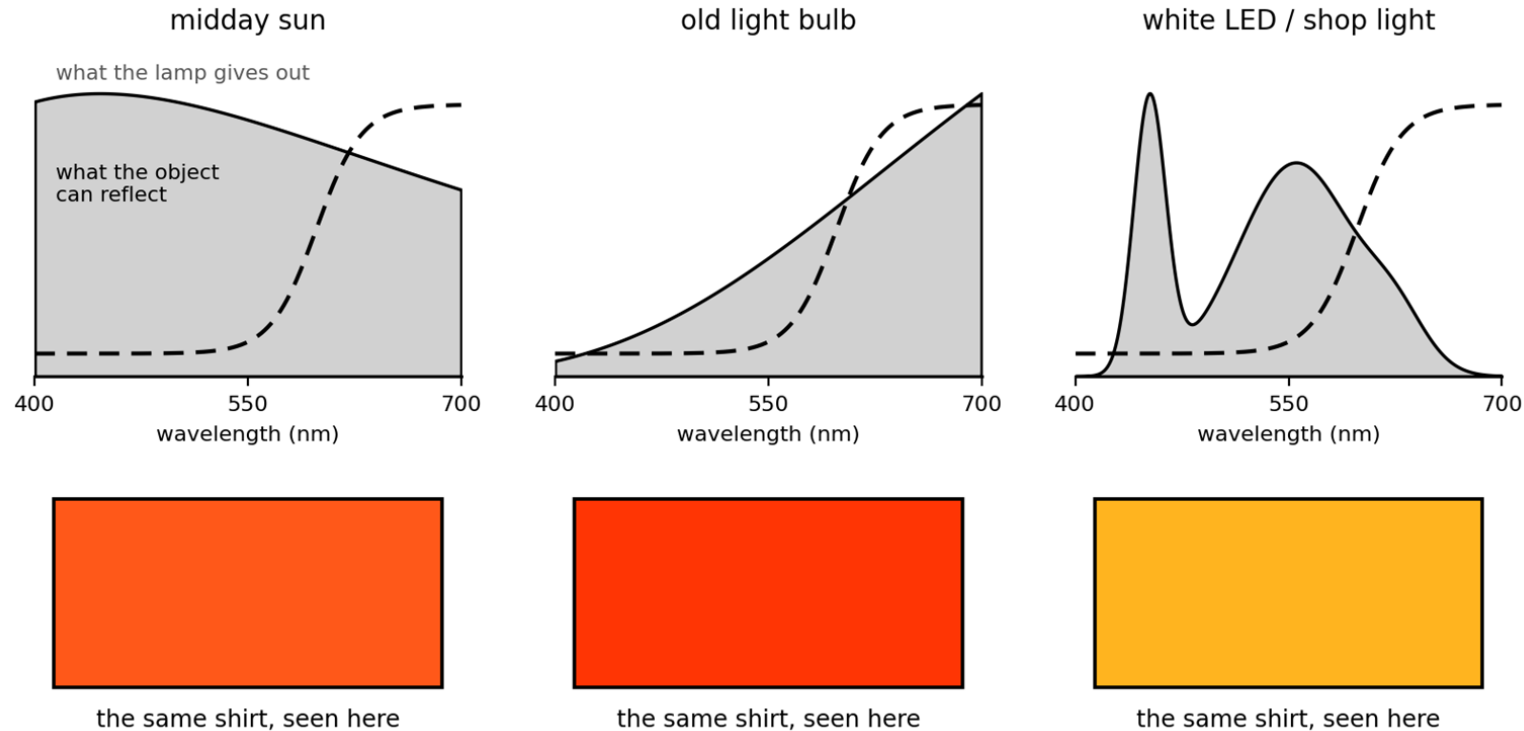


nothing bright is coming in, so that same faint
reflection is all your eye has to work with

Glass reflects about 4% of the light that hits it, always — day and night alike. During the day the scene outside is thousands of times brighter than you are, so that 4% is completely drowned out. At night nothing bright is coming in, so your faint reflection is the brightest thing your eye has to work with, and the window behaves like a mirror.

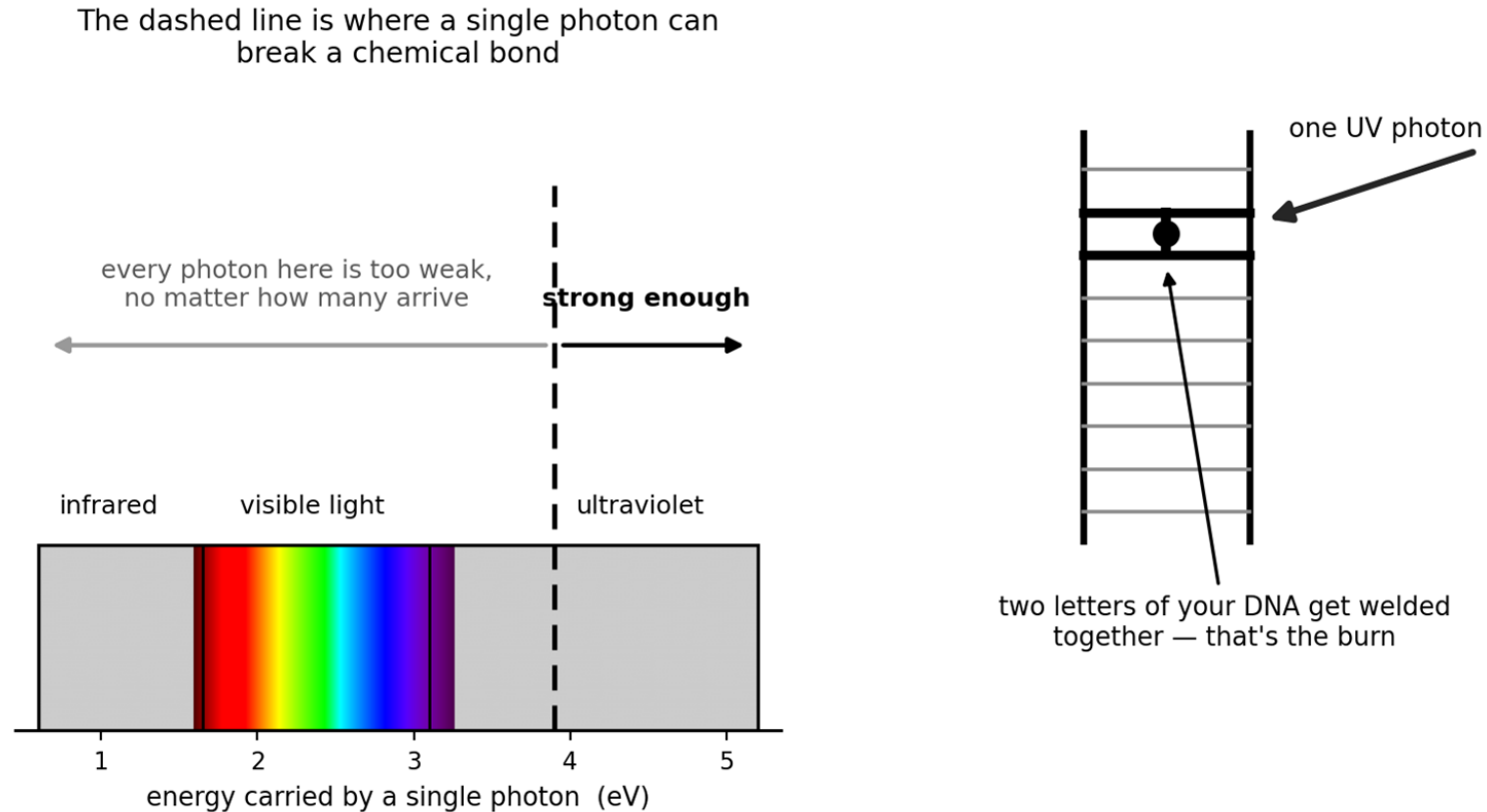
Why does an object's colour change under different lighting?

One object, three lamps — the light you get back is the lamp's spectrum times the object's



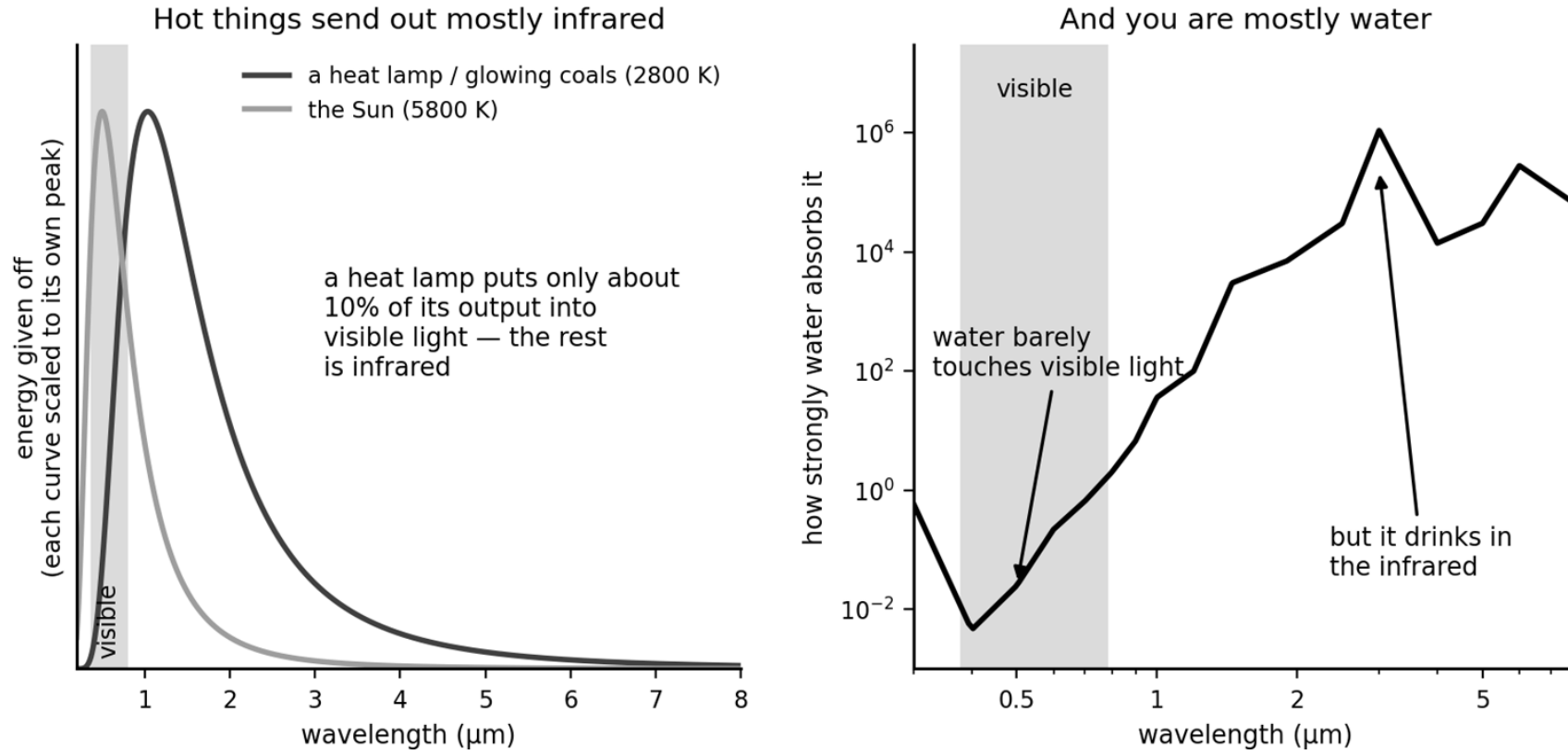
An object has no colour of its own — it only has a rule for what fraction of each wavelength it sends back. What reaches your eye is that rule multiplied by whatever the lamp happens to be putting out. A lamp that is weak in deep red has no red for a red shirt to reflect, so the shirt drifts toward orange or brown.

Why does ultraviolet burn you when you can't even see it?



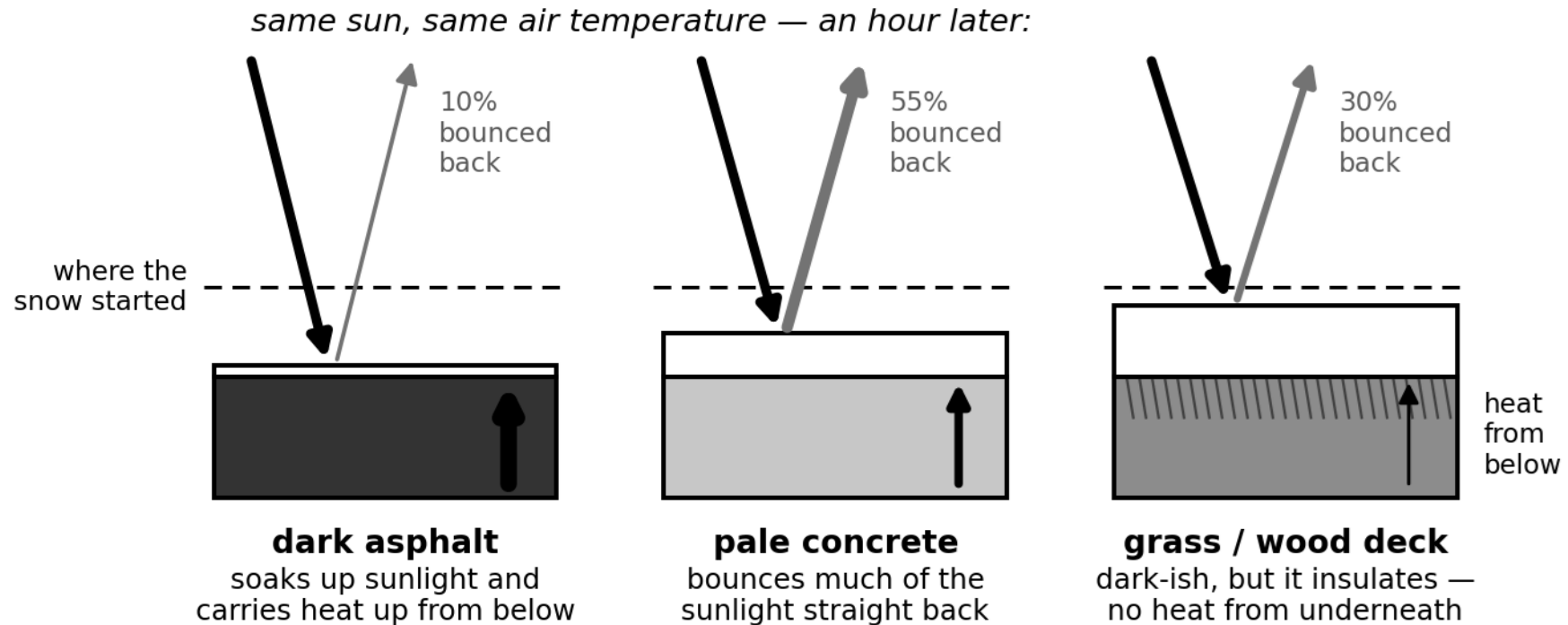
Damage is done one photon at a time, and brightness doesn't help. A single UV photon carries enough energy to break a chemical bond and weld two letters of your DNA together; a visible photon does not, no matter how many of them arrive. It is a hammer problem — a thousand gentle taps will not do what one hard strike does.

If visible light is more energetic, why does infrared heat better?



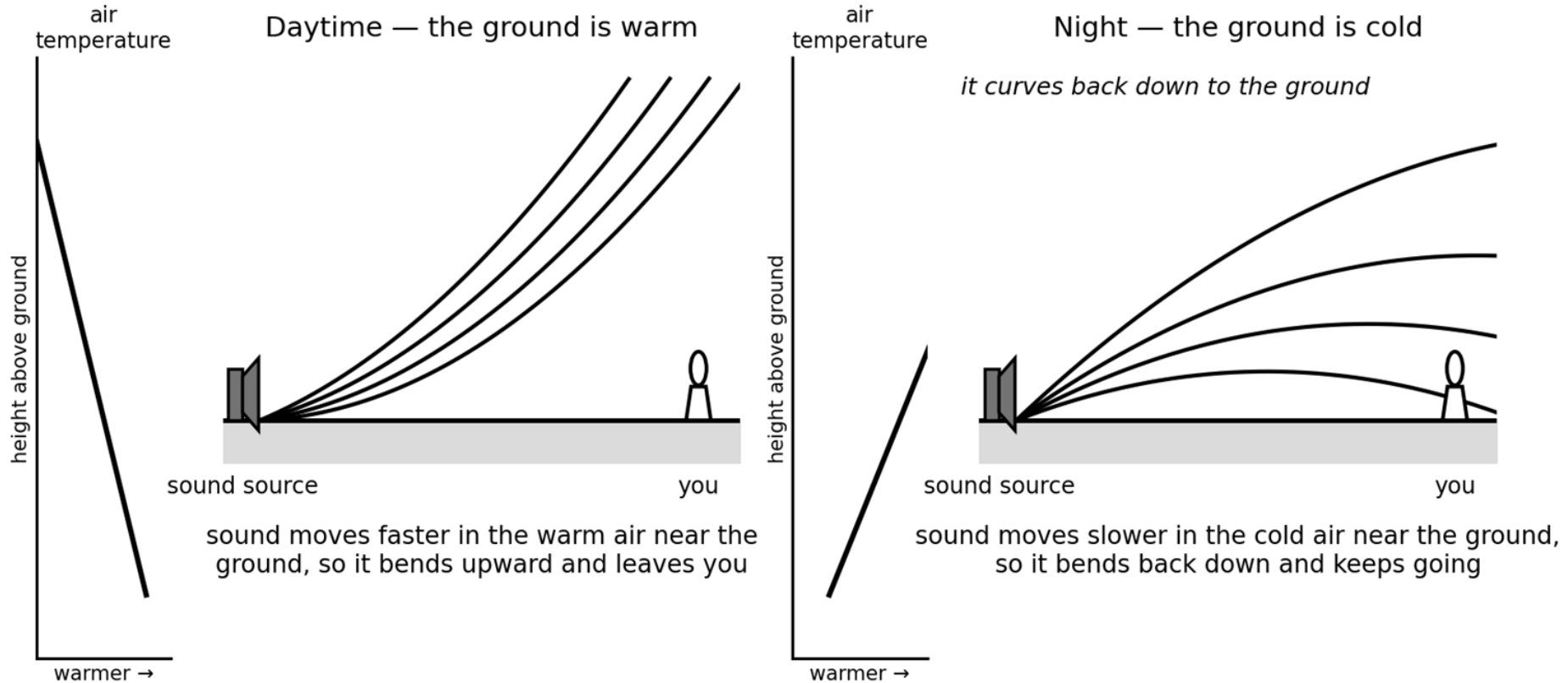
Per photon, visible light does win. But heating is about total energy absorbed, and two things swing it to infrared: anything warm radiates overwhelmingly in the IR, and you are mostly water, which absorbs infrared thousands of times more strongly than visible light. Visible light largely reflects off you or passes through; infrared gets stopped in your skin and becomes heat.

Why does snow melt faster on some surfaces at the same temperature?



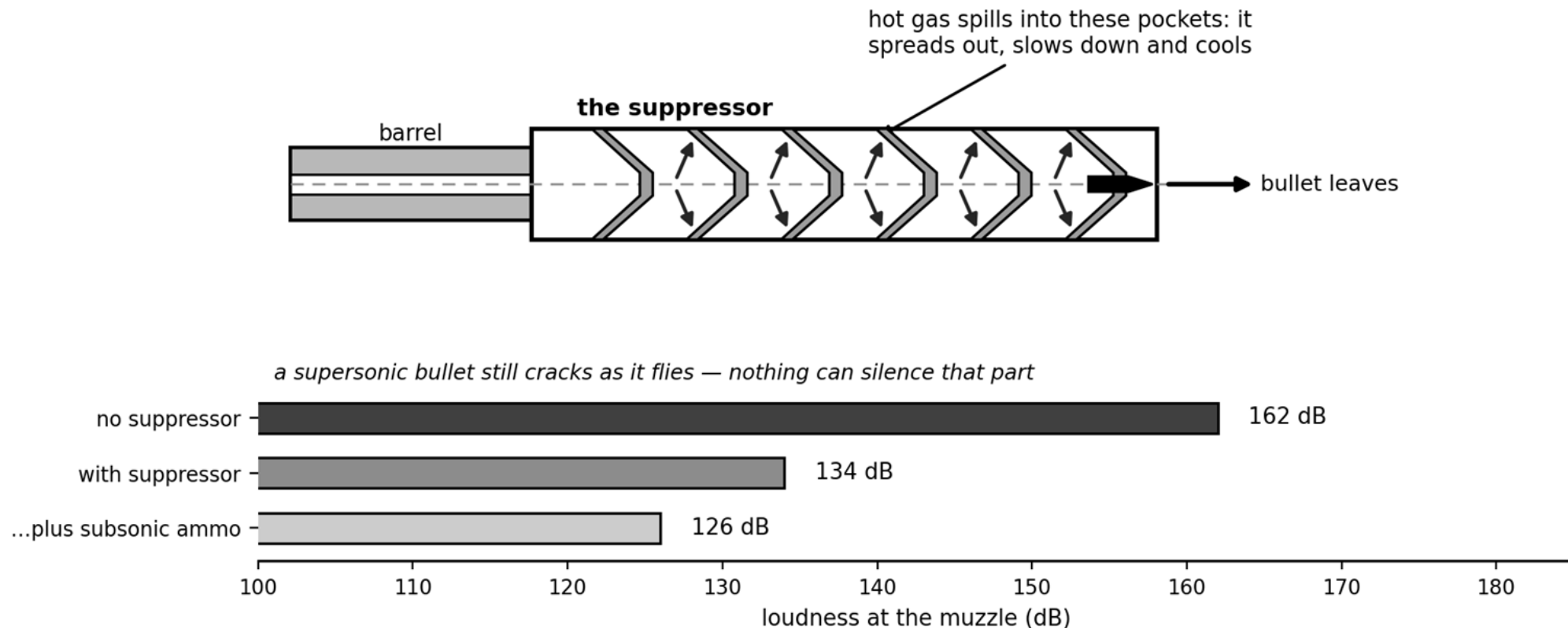
The air temperature is the same; three other things are not. How much sunlight the surface absorbs (dark asphalt takes nearly all of it, pale concrete throws half back), how well it conducts heat up from the warmer ground below, and how much heat it has stored. Grass and wood insulate, so the snow just sits there — asphalt both soaks up sun and feeds heat from underneath.

Why does sound travel farther at night?



Sound moves faster through warmer air, and it always bends toward the slower side. By day the ground is warm, so sound down low speeds up and curves upward — straight over your head. At night the ground cools first, so the lowest air is the slowest and sound curves back down, getting returned to the ground again and again instead of escaping.

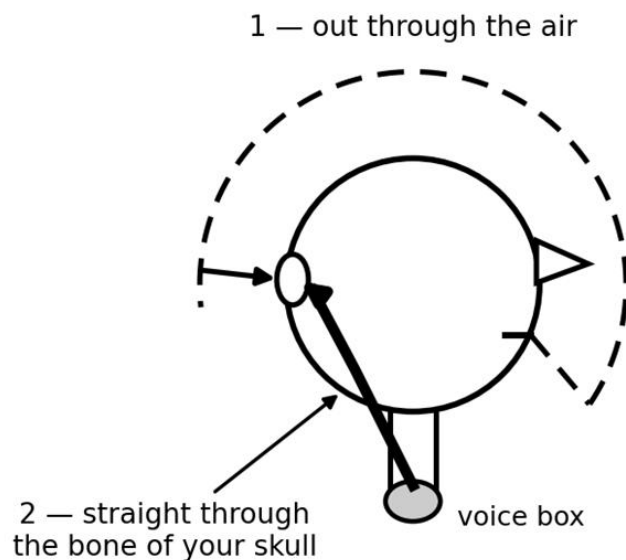
How does a suppressor actually work?



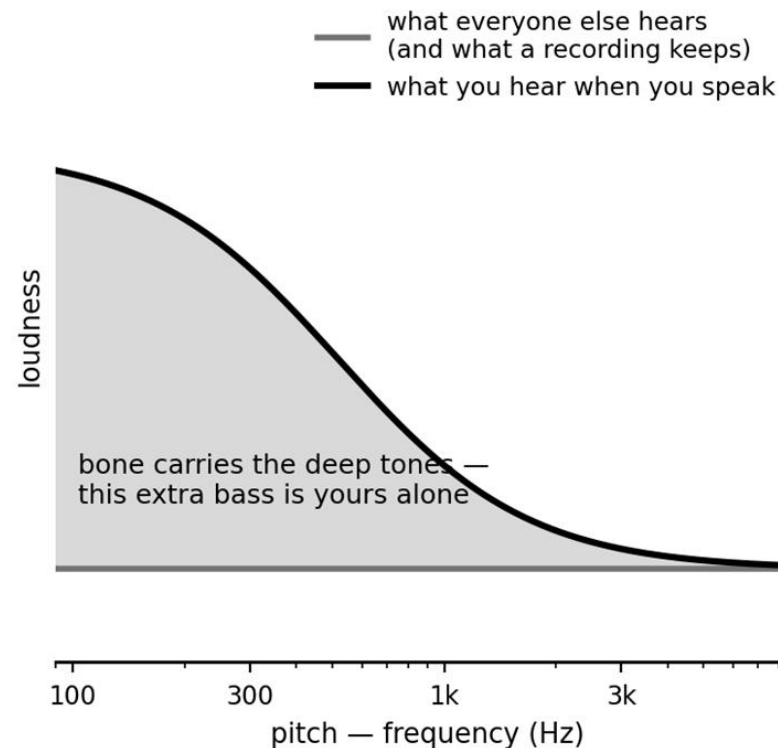
Most of the noise is the blast of hot, high-pressure gas bursting out of the muzzle all at once. A suppressor gives that gas a series of chambers to expand into first, so it spreads out, slows and cools before it ever reaches open air — typically 25–30 dB quieter. What no suppressor can remove is the crack of a supersonic bullet, which is why subsonic ammunition is quieter still.

Why does your voice sound wrong when you hear it played back?

Your voice reaches your ear two ways



So the recording sounds thin and high to you



When you speak, sound reaches your ears two ways: through the air, and directly through the bones of your skull. Bone carries low frequencies well, so the voice you hear live comes with extra bass that nobody else has ever heard. A recording captures only the air path — so it sounds thin and higher-pitched to you, and completely normal to everyone else.